Rishik Thammisetty

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Summary

Pre-final year Electrical and Electronics Engineering undergraduate at NIT Andhra Pradesh with interests in Machine Learning, VLSI, and Electronic Design Automation. Currently working as a Research Intern at IIT Bhilai on deep learning for image inpainting while developing physics-informed AI solutions for semiconductor applications.

Education

National Institute of Technology Andhra Pradesh — Tadepalligudem, India
B.Tech. in Electrical and Electronics Engineering; CGPA: 8.35/10 (Current) — ***Aug 2025 – Present***
Minor Degree in AI and Robotics — ***Aug 2025 – Present***

Technical Skills

Languages: C, Python, MATLAB, Verilog
ML Frameworks & Tools: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, SHAP
Core Domains: Machine Learning, Deep Learning, Image Processing, VLSI Design, EDA
Other Tools: Power BI, Xilinx VIVADO, Web Development, ESP32, Embedded Systems

Professional Experience

Indian Institute of Technology (IIT) Bhilai — **May 2026 – Present**
Research Intern
- Designed and developed a novel deep learning architecture for image inpainting to reconstruct missing and corrupted image regions with high visual fidelity.
- Investigated advanced convolutional and attention-based mechanisms to improve reconstruction quality, robustness, and computational efficiency.
- Conducted experimental evaluations using standard computer vision metrics, while exploring FPGA-based implementations to enable efficient hardware acceleration for real-time applications.

National Institute of Technology Andhra Pradesh — **Apr 2025 – Present**
Volunteer – E-Yantra Club
- Coordinated technical events and supported hands-on training sessions.
- Developed leadership, teamwork, and problem-solving skills through collaborative club activities.

Academic Projects

Deep Learning for Realistic Image Inpainting | Github — **May 2026 – Present**
- Designed and developed a novel deep learning architecture for image inpainting to reconstruct missing or corrupted image regions with high visual fidelity, utilizing convolutional and attention-based mechanisms.
- Conducted comprehensive experimental evaluations using standard computer vision metrics, optimizing the architecture for computational efficiency and real-time hardware acceleration via FPGA implementations.

Machine Learning-Assisted Static Timing Analysis (ML-STA) | Github — **May 2026 – Present**
- Developing an ML-assisted Static Timing Analysis framework to predict critical path delays, identify timing bottlenecks, and detect violations in digital circuits to accelerate design closure.
- Designing an end-to-end circuit analysis pipeline to extract high-dimensional timing features from gate-level netlists and timing graphs, reducing overall analysis complexity within EDA workflows.

Physics-Informed ML for CMOS Gate Delay Prediction | Github — **Jan 2026 – Present**
- Developed a physics-informed deep learning framework for 45nm CMOS gate delay prediction by embedding fundamental semiconductor physics principles directly into PyTorch-based neural networks.
- Performed explainability analysis using SHAP to verify physically meaningful decision-making, ensuring the model accurately prioritizes critical device and operating parameters to enhance interpretability.

EV Battery Health Prediction | Github — **Jan 2026 – Apr 2026**
- Developed an integrated ML-based system to predict State of Health (SoH) using State of Charge (SoC), Remaining Useful Life (RUL), and automated anomaly detection as supporting features.
- Applied robust data preprocessing, feature engineering, and regression algorithms to improve battery degradation prediction accuracy under dynamically varying operating conditions.

Relevant Coursework & Professional Development

- **NVIDIA Deep Learning Institute**: Deep Learning Workshop & Natural Language Processing (NLP) Workshop
- **NPTEL**: Design and Analysis of VLSI Subsystems (Elite Certified)
- **Advanced Domains**: Machine Learning and Electronic Design Automation (EDA) Methodologies